AVATAR · early surgery vs watchful waiting in asymptomatic severe aortic stenosis · outcome: all-cause death

RoB 2 asks up to 22 questions in 5 domains. Results describe risk of bias.

Y yes · PY / PN probably yes / no · N no · NI no information ▨ not needed (RoB 2 branching)

Evidence types: outlined yellow = direct support · grey = context · dashed = inference (no quote)

Sources stored 7 sources stored · 4 cited below

- **A · Circulation report · 2022** — fingerprint #592b9a
- **B · Eur Heart J report · 2024** — fingerprint #a890f7
- **C · ClinicalTrials.gov · 2015** — fingerprint #d32e23
- **D · Protocol amendment · 2021** — fingerprint #ba0a93

fingerprint = unique ID of the stored, unchangeable copy every quote is checked against

Domain & question	The sources say	AI model answers (answer + justification)	ARBITER (rule-based software, no AI)	Result
Bias arising from the randomisation process Did differences between the groups at baseline suggest a problem with randomisation? 1.3	**B · p 5** "Table 2 Selected baseline characteristics ..." **B · p 5** "Diabetes mellitus 14 (17.9%) 23 (29.1%) .07 ..."	**No** Baseline differences compatible with chance.	✓ 2 quotes found in the stored copies 1.1 Y · 1.2 Y · 1.3 N No randomisation concern.	**Low**
Bias due to deviations from intended interventions Did the trial itself cause departures from the intended treatment? 2.3	**B · p 6** "3-year follow-up ... available for all patients. ..." **D · p 2** "delayed surgery ... due to COVID -19 ..." **A · p 4** "referred ... onset of AS-related symptoms ..." **B · p 9** "physical visits ... substituted by phone follow-up. ..."	**Probably no** Changes followed the protocol or arose from the pandemic. *Every quote is checked*	✓ 4 quotes found in the stored copies 2.1 Y · 2.2 Y · 2.3 PN · 2.4 · 2.5 · 2.6 Y · 2.7 No trial-driven departures; appropriate analysis.	**Low**
Bias due to missing outcome data Were outcome data available for nearly all participants? 3.1	**B · p 6** "All patients were included in the analysis. ... One patient ... lost to follow-up ... censored ..."	**Yes** 1 of 157 (0.6%) incomplete.	✓ 1 quote found in the stored copies 3.1 Y · 3.2 · 3.3 · 3.4 Nearly complete data.	**Low**
Bias in measurement of the outcome Did assessors know the treatment? Could that affect how deaths were recorded? 4.3, 4.4	**B · p 3** "members were not blinded to ... the treatment allocation during events review. ..."	**Yes** **Probably no** *Death involves no judgement.*	✓ 1 quote found in the stored copies 4.1 PN · 4.2 PN · 4.3 Y · 4.4 PN · 4.5 Objective death count.	**Low**
Bias in selection of the reported result Was the analysis plan fixed before the results were known? 5.1	**C · p 1** "all cause death, ... (MACE) including: ..." **A · p 4** "Prespecified secondary ... Time to death ..." **B · p 5** "all analyses ... intention-to-treat principle, ..." **D · p 1** "-Amendment to surgical delays- ..."	**No information** No dated plan lists this analysis; death appears inside a combined registry outcome. *Sources silent*	✓ 4 quotes found in the stored copies 5.1 NI · 5.2 PN · 5.3 PN Plan not shown fixed in advance; no sign of selective reporting → Some concerns	**Some concerns**

ARBITER computes ratings

Weighed against: Diabetes was less common in early surgery (17.9% vs 29.1%), ... but P=.07 is compatible with chance

Weighed against: pandemic-related follow-up substitution by phone and lengthened AVR delays could have increased clinical risk. ... not the trial context ... no deaths occurred while awaiting scheduled surgery

? Open point: Reasons for declining surgery are not described in detail.

? Open point: Whether the published protocol listed time to all-cause death and its Cox analysis, and its date relative to unblinded data.

Overall Any domain with Some concerns → overall Some concerns (no domain High) — **Some concerns**

This trial: 15 questions answered · 7 not needed · 16 quotes, all found in the stored sources

All of this is saved in one audit file (.arbiter.zip), checked by an independent verifier: verified.

Published review: Low · Both expert adjudicators: Some concerns